

Dynamics of Equity Index Volatility:

Forecasting Performance of GARCH-Type Models versus Implied
Volatility and Implications for the Volatility Smile

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Table des matières

0.1	Data Cleaning and Preprocessing	3
0.1.1	Data Construction	3
0.1.2	Data Cleaning	4
0.1.3	Outlier Treatment	4
0.2	Realized Volatility Estimation	4
0.2.1	Definition	4
0.2.2	Rolling Windows	5
0.2.3	Interpretation	5
0.2.4	Stylized Facts	5
0.3	Implied Volatility	6
0.3.1	Definition	6
0.3.2	Data Construction	6
0.3.3	Implied vs Realized Volatility	6
0.3.4	Volatility Risk Premium	7
0.3.5	Economic Interpretation	7
0.3.6	Link with Volatility Modeling	7
0.4	GARCH-Type Volatility Models	8
0.4.1	Motivation	8
0.4.2	Baseline GARCH Model	8
0.4.3	Student-t Innovations	8
0.4.4	EGARCH Model	8
0.4.5	GJR-GARCH Model	9
0.4.6	Model Specifications	9
0.4.7	Training and Testing Framework	9
0.4.8	Volatility Forecasting	9
0.4.9	Economic Interpretation	10
0.5	Forecast Evaluation	10
0.5.1	Evaluation Framework	10
0.5.2	Empirical Results	10
0.5.3	Diebold-Mariano Statistical Tests	11
0.5.4	Discussion	13
0.5.5	Economic Interpretation	14
0.5.6	Key Takeaway	14
0.6	Risk Management and Tail-Risk Backtesting	14
0.6.1	Value-at-Risk Framework	14
0.6.2	Kupiec Unconditional Coverage Test	14
0.6.3	Christoffersen Independence Test	15
0.6.4	Expected Shortfall Analysis	15
0.6.5	Empirical Backtesting Results	15

0.6.6	Economic Interpretation	16
0.7	Conclusion	16
0.8	Limitations and Extensions	17
0.8.1	Limitations	17
0.8.2	Potential Extensions	17

0.1 Data Cleaning and Preprocessing

I consider daily price data for the S&P 500 over the period from January 2, 2015 to May 4, 2026. The dataset contains a total of 2,850 observations.

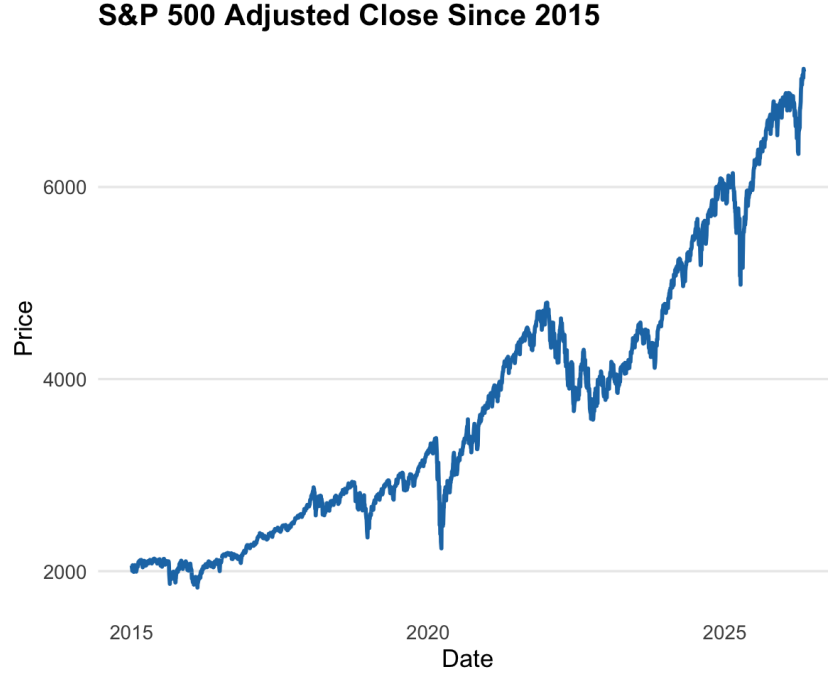


FIGURE 1 – Evolution of S&P 500 (Adjusted close price)

0.1.1 Data Construction

Let P_t denote the closing price at time t . I compute continuously compounded returns as :

$$r_t = \log(P_t) - \log(P_{t-1}) \quad (1)$$

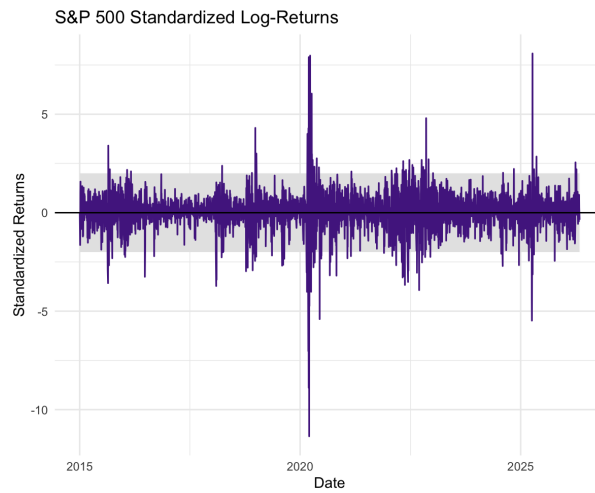


FIGURE 2 – log returns

This transformation ensures stationarity and is standard in financial econometrics. This stationarity is verified using ADF test :

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Augmented Dickey-Fuller Test

data: log_returns_std
Dickey-Fuller = -14.472, Lag order = 14, p-value = 0.01
alternative hypothesis: stationary

```

0.1.2 Data Cleaning

The dataset is carefully cleaned to ensure consistency and robustness :

- Missing values are removed.
- Duplicate timestamps are checked and eliminated.
- The time index is verified to ensure proper chronological ordering.

After cleaning, the dataset exhibits the following properties :

- Start date : January 2, 2015
- End date : May 4, 2026
- Number of observations : 2,850
- Missing values : 0
- Duplicate timestamps : 0

0.1.3 Outlier Treatment

Extreme returns are not removed, as they contain important information about market stress and volatility clustering. Instead, I flag extreme observations using a high quantile threshold :

$$|r_t| > q_{0.999} \quad (2)$$

where $q_{0.999}$ denotes the 99.9th percentile of absolute returns.

This approach preserves tail behavior while allowing us to monitor extreme events.

0.2 Realized Volatility Estimation

To measure time-varying market risk, I construct realized volatility using rolling windows of different lengths.

0.2.1 Definition

For a given window size w , realized volatility is defined as :

$$\sigma_t^{(w)} = \sqrt{252 \cdot \frac{1}{w} \sum_{i=1}^w r_{t-i}^2} \quad (3)$$

where :

- r_t denotes daily log-returns,
- w is the rolling window length,
- the factor 252 annualizes the volatility.

0.2.2 Rolling Windows

I compute realized volatility using multiple horizons :

- $w = 5$ days (weekly horizon),
- $w = 10$ days (bi-weekly horizon),
- $w = 21$ days (monthly horizon),
- $w = 63$ days (quarterly horizon).

This multi-scale approach allows us to capture both short-term and long-term volatility dynamics.

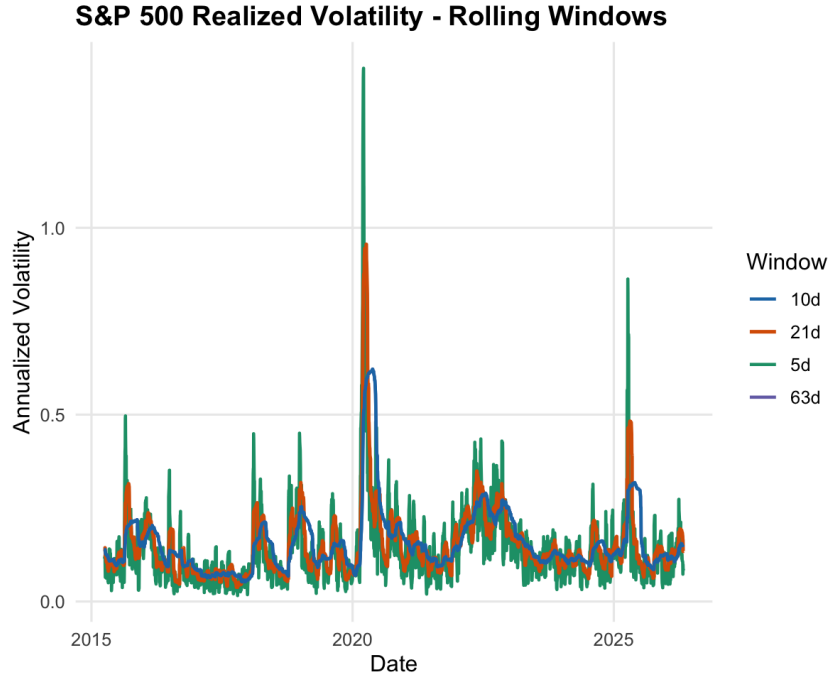


FIGURE 3 – Realized volatility computed for different time windows

0.2.3 Interpretation

Shorter windows react more quickly to market shocks but are noisier, while longer windows provide smoother estimates at the cost of slower responsiveness.

This trade-off is central in volatility modeling and will be further explored when evaluating forecasting performance.

0.2.4 Stylized Facts

Preliminary inspection of realized volatility reveals :

- Volatility clustering : periods of high volatility tend to persist.
- Time-varying variance : volatility is clearly not constant over time.
- Asymmetry : large negative returns are often associated with spikes in volatility.

These empirical features motivate the use of conditional heteroskedasticity models such as GARCH and its extensions.

0.3 Implied Volatility

0.3.1 Definition

Implied volatility represents the market's expectation of future volatility embedded in option prices. Unlike realized volatility, which is computed from historical returns, implied volatility is forward-looking and reflects investor sentiment and risk pricing.

In this study, I use the VIX index as a proxy for implied volatility. The VIX is constructed from S&P 500 index options and reflects the market's expectation of volatility over the next 30 days.

0.3.2 Data Construction

I retrieve daily observations of the VIX index over the same sample period as the S&P 500 data. Since the VIX is expressed in percentage terms, I convert it into decimal form :

$$IV_t = \frac{VIX_t}{100} \quad (4)$$

The implied volatility series is then aligned with the realized volatility dataset by matching trading dates and removing missing observations.

0.3.3 Implied vs Realized Volatility

A key objective of this project is to compare implied volatility with realized volatility. In particular, I focus on the 21-day realized volatility as a proxy for future volatility over a horizon comparable to the VIX.

Empirically, implied volatility tends to exceed realized volatility on average, reflecting a compensation demanded by investors for bearing volatility risk.

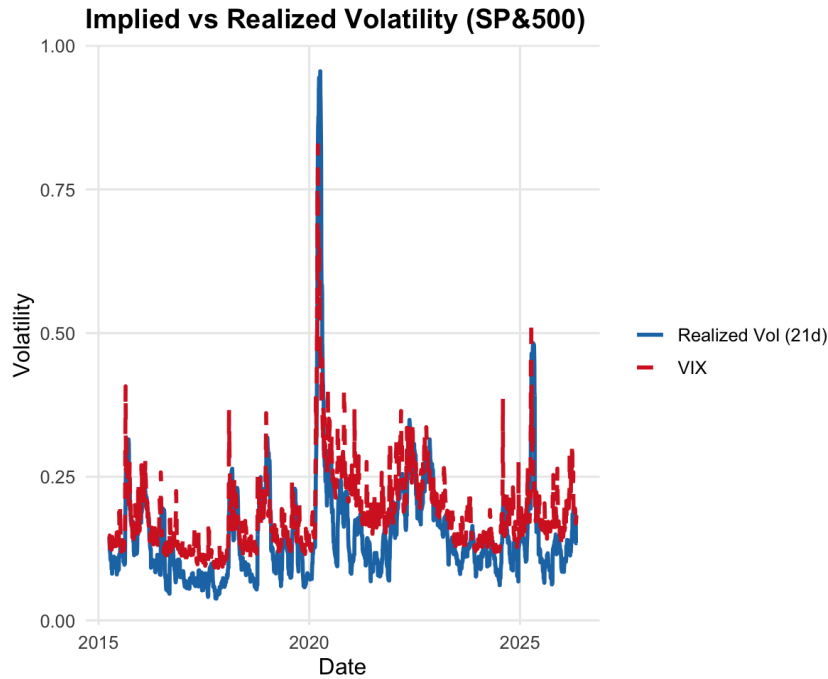


FIGURE 4 – Implied vs Realized volatility

0.3.4 Volatility Risk Premium

I define the volatility risk premium (VRP) as the difference between implied and realized volatility :

$$\text{VRP}_t = IV_t - \sigma_t^{(21)} \quad (5)$$

where $\sigma_t^{(21)}$ denotes the 21-day realized volatility.

The VRP captures the extent to which the market overestimates or underestimates future volatility :

- $\text{VRP}_t > 0$: implied volatility exceeds realized volatility, indicating a positive risk premium.
- $\text{VRP}_t < 0$: realized volatility exceeds implied volatility, often during periods of market stress.

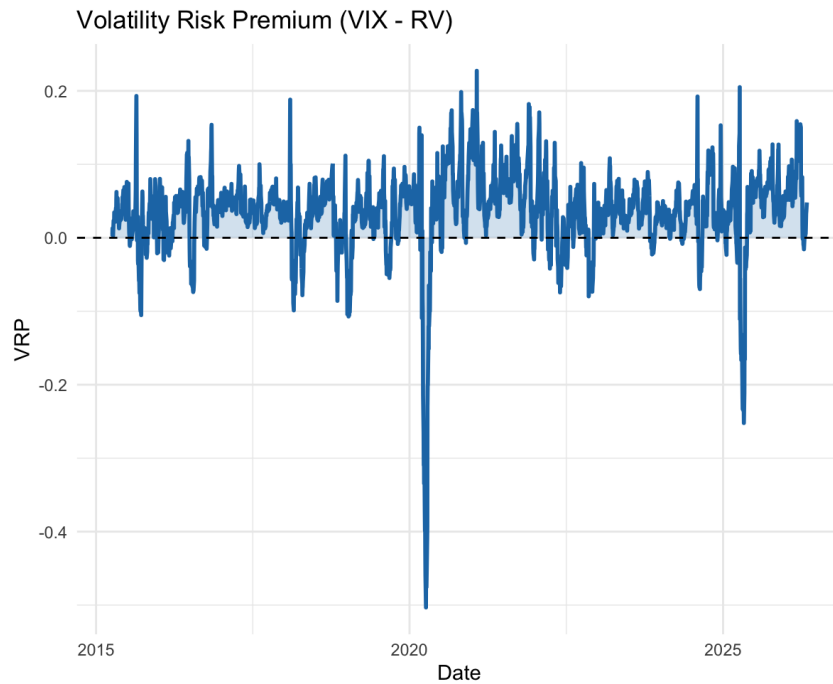


FIGURE 5 – Volatility risk premium

0.3.5 Economic Interpretation

The presence of a positive volatility risk premium is a well-documented phenomenon in financial markets. It reflects investors' aversion to volatility and their willingness to pay for protection against downside risk.

Analyzing the dynamics of the VRP provides insights into market expectations, risk perception, and potential mispricing, which are central to volatility modeling and forecasting.

0.3.6 Link with Volatility Modeling

Implied volatility serves as a natural benchmark for model-based forecasts. In the following sections, we will compare the predictive performance of GARCH-type models

against implied volatility to assess whether statistical models can outperform the market's expectations.

0.4 GARCH-Type Volatility Models

0.4.1 Motivation

Financial returns typically exhibit volatility clustering, meaning that periods of high volatility tend to be followed by periods of high volatility, and periods of low volatility tend to persist. This empirical regularity motivates the use of conditional heteroskedasticity models.

In this section, I estimate several GARCH-type models in order to forecast future volatility and compare model-based forecasts with realized and implied volatility benchmarks.

0.4.2 Baseline GARCH Model

The standard GARCH(1, 1) model specifies returns as :

$$r_t = \mu + \varepsilon_t \quad (6)$$

$$\varepsilon_t = \sigma_t z_t \quad (7)$$

where z_t is an i.i.d. motion process and σ_t^2 is the conditional variance.

The conditional variance follows :

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8)$$

where :

- ω is the long-run variance component,
- α measures the short-term reaction to market shocks,
- β captures volatility persistence.

A high value of $\alpha + \beta$ indicates strong persistence in volatility.

0.4.3 Student-t Innovations

Since financial returns often exhibit fat tails, I also estimate GARCH models with Student-t motions rather than Gaussian motions. This allows the model to better capture extreme market movements.

0.4.4 EGARCH Model

To account for asymmetric volatility responses, I estimate an Exponential GARCH model. The EGARCH(1, 1) specification models the logarithm of conditional variance :

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \alpha (|z_{t-1}| - \mathbb{E}[|z_{t-1}|]) + \gamma z_{t-1} \quad (9)$$

The parameter γ captures the leverage effect. A negative value indicates that negative returns increase future volatility more than positive returns of the same magnitude.

0.4.5 GJR-GARCH Model

I also estimate a GJR-GARCH model, which introduces an explicit asymmetry term :

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \mathbb{I}_{\{\varepsilon_{t-1} < 0\}} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (10)$$

where $\mathbb{I}_{\{\varepsilon_{t-1} < 0\}}$ is an indicator function equal to one when the previous shock is negative.

If $\gamma > 0$, negative shocks have a larger impact on conditional volatility than positive shocks.

0.4.6 Model Specifications

I estimate the following models :

- GARCH(1, 1) with Student-t innovations,
- EGARCH(1, 1) with Student-t innovations,
- GJR-GARCH(1, 1) with Student-t innovations.

This model set allows us to compare the impact of fat-tailed innovations and asymmetric volatility dynamics.

0.4.7 Training and Testing Framework

The sample is split into an in-sample training period and an out-of-sample testing period. Models are initially estimated on the training sample ending on December 31, 2021. Forecasting performance is then evaluated on the out-of-sample period starting in January 2022.

To avoid look-ahead bias, forecasts are generated using a rolling estimation procedure. The model is re-estimated every 21 trading days using a moving window.

0.4.8 Volatility Forecasting

For each model, I generate one-step-ahead volatility forecasts. Since the GARCH models are estimated on daily returns, the resulting daily volatility forecasts are annualized as :

$$\hat{\sigma}_{t,\text{ann}} = \sqrt{252} \cdot \hat{\sigma}_{t,\text{daily}} \quad (11)$$

The forecasts are then compared with :

- 21-day realized volatility,
- VIX implied volatility,
- alternative GARCH-type model forecasts.

GARCH Forecasts vs Realized Volatility

Annualized volatility comparison: real. volatility, VIX and GARCH-family forecasts

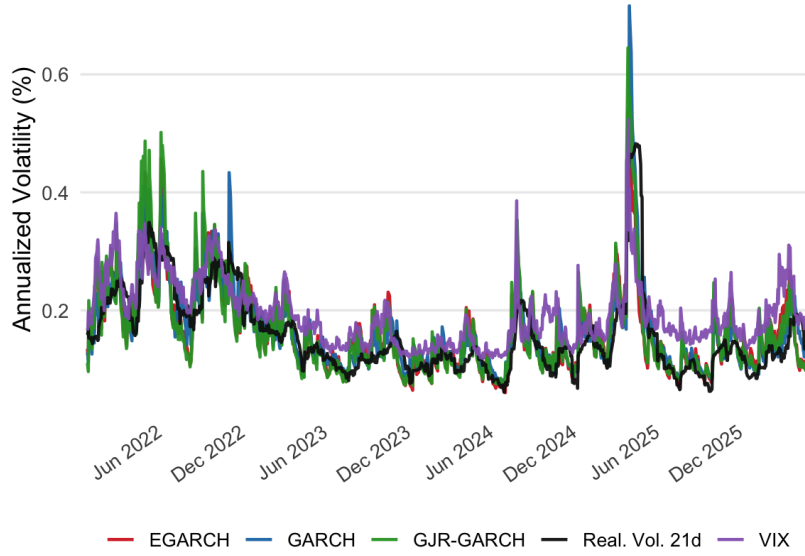


FIGURE 6 – Garch forecasts vs realized volatility

0.4.9 Economic Interpretation

The purpose of this modeling framework is not only to identify the model with the best statistical fit, but also to assess whether econometric volatility forecasts can compete with the market-implied benchmark.

In particular, the comparison between GARCH-type forecasts and the VIX helps address the central research question of this project :

Can statistical volatility models outperform market-implied volatility expectations?

0.5 Forecast Evaluation

0.5.1 Evaluation Framework

To assess the predictive performance of volatility models, I compare model-based forecasts with realized volatility using several loss functions :

- Mean Squared Error (MSE),
- Mean Absolute Error (MAE),
- QLIKE loss function.

The QLIKE loss is particularly relevant in volatility forecasting, as it is robust to measurement error in realized volatility and directly evaluates variance forecasts.

0.5.2 Empirical Results

Table 1 reports the out-of-sample forecasting performance of the models.

Model	MSE	MAE	QLIKE	Relative QLIKE
GARCH (Student-t)	0.00139	0.02475	-2.77466	1.049
EGARCH (Student-t)	0.00243	0.03533	-2.70404	1.022
GJR-GARCH (Student-t)	0.00271	0.03633	-2.70216	1.022
VIX	0.00360	0.04790	-2.64506	1.000

TABLE 1 – Out-of-sample volatility forecast evaluation

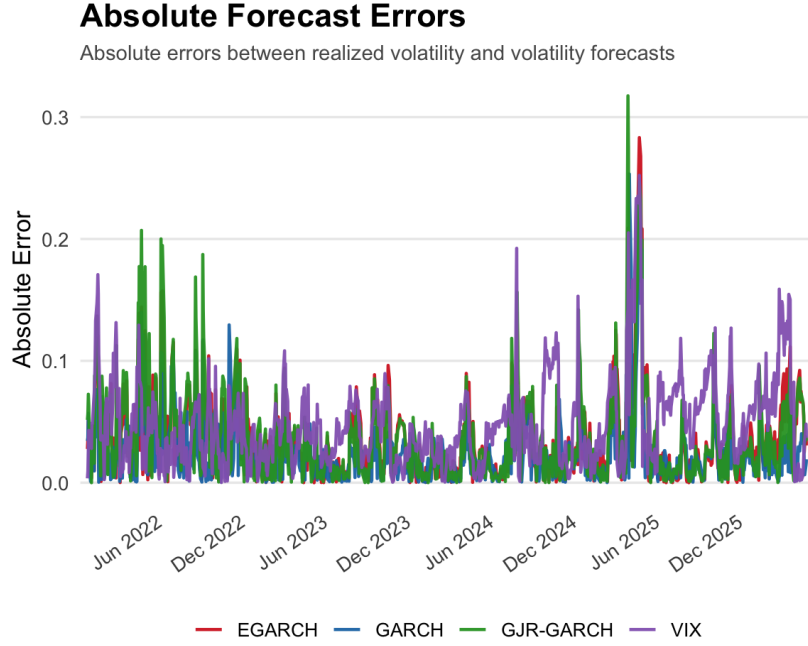


FIGURE 7 – Absolute forecast errors

0.5.3 Diebold-Mariano Statistical Tests

While standard forecasting metrics such as MSE, MAE and QLIKE provide a ranking of model performance, they do not indicate whether the observed differences are statistically significant. To formally compare competing volatility forecasts, Diebold-Mariano (DM) tests were implemented.

The Diebold-Mariano framework evaluates the null hypothesis of equal predictive accuracy between two forecasting models. Let :

$$d_t = L(e_{1,t}) - L(e_{2,t})$$

where :

- $L(\cdot)$ denotes a forecasting loss function,
- $e_{1,t}$ and $e_{2,t}$ are the forecast errors of the two competing models.

Under the null hypothesis :

$$H_0 : \mathbb{E}[d_t] = 0$$

meaning that both models have equal predictive ability.

The DM statistic is asymptotically normally distributed :

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}$$

where $\bar{d}$ is the average loss differential.

The tests were conducted using the MSE, MAE and QLIKE loss functions for the following comparisons :

- VIX vs GARCH Student-t,
- VIX vs EGARCH Student-t,
- GARCH Student-t vs EGARCH Student-t,
- EGARCH Student-t vs GJR-GARCH Student-t.

TABLE 2 – Diebold-Mariano Test Results

Model 1	Model 2	Loss	DM Statistic	p-value	Preferred Model
VIX	GARCH	MSE	12.82	< 0.001	GARCH
VIX	GARCH	MAE	20.91	< 0.001	GARCH
VIX	GARCH	QLIKE	19.56	< 0.001	GARCH
VIX	EGARCH	MSE	7.82	< 0.001	EGARCH
VIX	EGARCH	MAE	11.10	< 0.001	EGARCH
VIX	EGARCH	QLIKE	6.94	< 0.001	EGARCH
GARCH	EGARCH	MSE	-8.16	< 0.001	GARCH
GARCH	EGARCH	MAE	-14.44	< 0.001	GARCH
GARCH	EGARCH	QLIKE	-12.56	< 0.001	GARCH
EGARCH	GJR-GARCH	MSE	-2.59	0.010	EGARCH
EGARCH	GJR-GARCH	MAE	-2.14	0.032	EGARCH
EGARCH	GJR-GARCH	QLIKE	-0.54	0.588	EGARCH

The Diebold-Mariano results confirm that the forecasting performance differences are statistically significant.

First, both GARCH and EGARCH significantly outperform the VIX benchmark across all loss functions. This indicates that econometric volatility models contain additional predictive information beyond implied volatility alone.

Second, the GARCH Student-t specification significantly outperforms the EGARCH model under MSE, MAE and QLIKE losses. Despite its simpler symmetric structure, the standard GARCH model delivers the strongest overall volatility forecasting accuracy in this empirical setting.

Finally, the comparison between EGARCH and GJR-GARCH reveals weaker statistical differences. Although EGARCH performs slightly better under MSE and MAE, the QLIKE comparison is not statistically significant.

Overall, these results demonstrate that the superiority of the selected volatility models is not only visual or descriptive, but also statistically robust according to formal predictive accuracy tests.

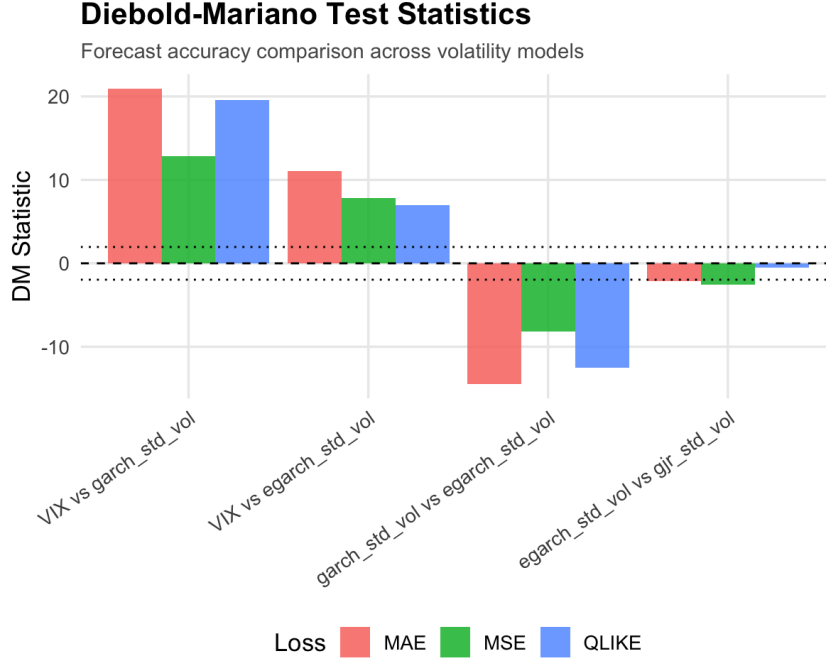


FIGURE 8 – Diebold-Mariano test

0.5.4 Discussion

Several important conclusions emerge from the empirical analysis.

First, all GARCH-family models outperform the VIX benchmark in terms of MSE, MAE and QLIKE loss functions. The Diebold-Mariano tests further confirm that these performance improvements are statistically significant, indicating that econometric volatility models contain predictive information beyond market-implied volatility alone.

Second, the standard GARCH(1,1) model with Student-t innovations delivers the best overall volatility forecasting performance. This result highlights the importance of modeling heavy-tailed return distributions when forecasting financial market volatility. Despite its relatively simple symmetric structure, the GARCH Student-t specification consistently dominates more complex asymmetric models in out-of-sample forecasting accuracy.

Third, although EGARCH and GJR-GARCH models explicitly incorporate leverage effects and asymmetric volatility dynamics, these extensions do not systematically improve volatility forecasting performance in this dataset.

0.5.5 Economic Interpretation

The comparison between model-based forecasts and implied volatility provides key insights into market efficiency.

While GARCH models improve statistical accuracy, the VIX remains a strong benchmark as it aggregates forward-looking information from option markets.

These results suggest that purely econometric models may capture historical dynamics effectively but may still lack the forward-looking information embedded in option prices.

0.5.6 Key Takeaway

Overall, the results indicate that :

- GARCH models outperform the VIX in terms of classical error metrics,
- the Student-t specification significantly improves performance,
- implied volatility remains difficult to outperform in variance-based loss functions.

This highlights the importance of combining statistical models with market-based information in volatility forecasting.

0.6 Risk Management and Tail-Risk Backtesting

0.6.1 Value-at-Risk Framework

While volatility forecasting is a central component of quantitative finance, risk management applications require evaluating whether volatility models are capable of accurately capturing extreme downside events. For this reason, the rolling GARCH forecasts were extended to a Value-at-Risk (VaR) framework.

For a confidence level α , the one-day ahead VaR corresponds to the loss threshold that should only be exceeded with probability α :

$$VaR_{\alpha,t+1} = \mu_{t+1} + \sigma_{t+1}q_{\alpha}$$

where :

- μ_{t+1} is the conditional mean forecast,
- σ_{t+1} is the conditional volatility forecast,
- q_{α} is the quantile of the assumed innovation distribution.

The analysis focuses on the 1% and 5% VaR levels using the rolling forecasts generated by the GARCH, EGARCH and GJR-GARCH models with Student-t innovations.

0.6.2 Kupiec Unconditional Coverage Test

To evaluate whether the VaR forecasts produce the correct number of violations, the Kupiec unconditional coverage test was implemented.

Let :

$$\hat{p} = \frac{x}{N}$$

where :

- x is the number of observed violations,
- N is the total number of forecasts.

The Kupiec likelihood-ratio statistic is defined as :

$$LR_{uc} = -2 \log \left(\frac{(1 - \alpha)^{N-x} \alpha^x}{(1 - \hat{p})^{N-x} \hat{p}^x} \right)$$

Under the null hypothesis of correct unconditional coverage, the statistic follows a $\chi^2(1)$ distribution.

A rejection of the null hypothesis indicates that the model underestimates or overestimates downside risk.

0.6.3 Christoffersen Independence Test

In addition to the number of violations, a reliable VaR model should also produce violations that are independently distributed over time.

The Christoffersen independence test evaluates whether violations occur randomly or cluster during periods of market stress.

The test compares :

- a restricted model assuming independent violations,
- an unrestricted first-order Markov process.

A rejection of the null hypothesis would indicate volatility clustering not properly captured by the model.

0.6.4 Expected Shortfall Analysis

Beyond VaR exceedances, the analysis also considers Expected Shortfall (ES), which measures the average realized loss conditional on a VaR violation :

$$ES_\alpha = \mathbb{E}[r_t \mid r_t < VaR_{\alpha,t}]$$

Expected Shortfall provides additional information regarding the severity of tail losses and complements the VaR backtesting framework.

0.6.5 Empirical Backtesting Results

Table 2 reports the empirical VaR backtesting results for the different GARCH-family specifications.

TABLE 3 – VaR Backtesting Results

Model	α	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	ES
GARCH	1%	16	1.47%	0.148	0.230	-2.696
GARCH	5%	85	7.79%	0.0001	0.237	-2.014
EGARCH	1%	16	1.47%	0.148	0.230	-2.676
EGARCH	5%	72	6.60%	0.021	0.704	-2.078
GJR-GARCH	1%	16	1.47%	0.148	0.490	-2.398
GJR-GARCH	5%	79	7.24%	0.001	0.414	-2.028

The empirical results show that all models provide statistically acceptable VaR forecasts at the 1% level, as the Kupiec and Christoffersen tests are not rejected.

However, at the 5% level, all models fail the Kupiec test, indicating that the observed number of violations exceeds the theoretical level implied by the models. This suggests that downside risk is underestimated during moderately stressed market conditions.

Among the competing specifications, the EGARCH model delivers the best overall performance at the 5% level, with the lowest violation rate and the highest Kupiec p-value.

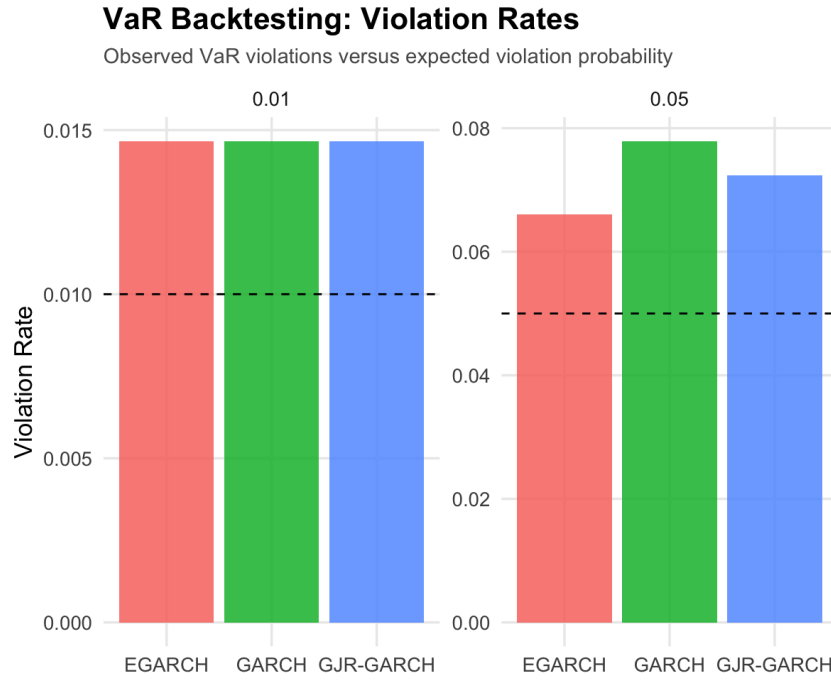


FIGURE 9 – Graph of violation rates for VAR 1% and 5%

0.6.6 Economic Interpretation

These results highlight an important distinction between volatility forecasting accuracy and tail-risk forecasting performance.

Although all GARCH-family models produce relatively accurate volatility forecasts, their ability to capture extreme downside events differs significantly. The superior performance of the EGARCH specification suggests that accounting for asymmetric volatility dynamics improves tail-risk estimation.

This result is economically consistent with the leverage effect observed in equity markets, where negative returns generate larger increases in future volatility than positive returns of the same magnitude.

Overall, the backtesting framework demonstrates that volatility models should not only be evaluated through forecasting loss functions such as MSE or QLIKE, but also through their capacity to capture market tail risk in a statistically robust manner.

0.7 Conclusion

This study investigates the ability of econometric models to forecast equity market volatility and compares their performance with market-implied expectations.

Using daily data on the S&P500, I construct realized volatility measures and evaluate several GARCH-type models, including GARCH, EGARCH, and GJR-GARCH specifications with Student-t innovations.

The empirical results show that :

- GARCH-type models outperform the VIX in terms of traditional error metrics such as MSE and MAE,
- accounting for fat tails via Student-t distributions significantly improves forecasting performance,

- more complex asymmetric models (EGARCH, GJR-GARCH) do not systematically outperform the standard GARCH specification.

However, when evaluated using the QLIKE loss function, implied volatility remains a strong benchmark, highlighting the difficulty of outperforming market-based forecasts.

Overall, the results suggest that while econometric models capture historical volatility dynamics effectively, they do not fully incorporate forward-looking information embedded in option prices.

This reinforces the view that combining statistical models with market-implied information is essential for accurate volatility forecasting.

0.8 Limitations and Extensions

While the results provide valuable insights, several limitations should be acknowledged.

0.8.1 Limitations

- The analysis relies on daily data, which may not fully capture high-frequency volatility dynamics.
- Realized volatility is approximated using rolling windows rather than high-frequency measures.
- The models are purely backward-looking and do not incorporate exogenous information.
- Only one asset is considered, limiting cross-sectional generalization.

0.8.2 Potential Extensions

Several extensions could significantly enhance the analysis.

High-Frequency Volatility Measures Using intraday data, one can compute more accurate volatility estimators such as realized variance, bipower variation, and jump variation. These measures allow for a decomposition of volatility into continuous and jump components.

Jump Detection Incorporating jump processes could improve volatility modeling, especially during periods of market stress.

Hybrid Models Combining GARCH-type models with implied volatility could lead to improved forecasts. For example, one could include the VIX as an exogenous variable in a GARCH-X framework.

Machine Learning Approaches Modern approaches such as LSTM networks or random forests could capture nonlinear patterns not accounted for by classical models.

Multi-Asset Analysis Extending the analysis to multiple asset classes (equities, FX, commodities) would improve the robustness and generality of the results.